CAPSTONE, PORTFOLIO & INTERVIEWS · LESSON 10 OF 10

A Realistic 12-Month Roadmap for a Beginner

Plan a year of learning in twelve steps, from Excel and SQL foundations through business intelligence, Python, statistics and machine learning, data engineering and cloud, GenAI and agents, to a portfolio and interview preparation. Each stage has weekly hours, a milestone, an assessment, and a plain statement of what employers generally expect.

Gap-fill lesson | ✦ 2026 | Beginner | ⏱ 90 min

By the end of this lesson you can

- Plan twelve months of learning with weekly hours and milestones
- Choose the order of topics and explain why it matters
- Set an assessment for each stage that proves the skill
- Set realistic expectations about what employers look for

Why a roadmap, and why realistic

A roadmap turns a vague wish ("I want to work with data") into a sequence of skills, each proven by a piece of work. The timing matters: a beginner who plans ten hours a week and finishes in twelve months has a real chance; a plan of thirty hours a week for three months usually stalls by the fourth week.

CASE STUDY **Two learners, one year**

Two learners start in January with the same goal. The first studies whenever motivated and skips the projects; by December she has watched many videos and has nothing to show. The second follows a roadmap of about ten hours a week, finishes a project each stage and keeps a record of each milestone. By December he has three published projects and a plan for interviews. The difference is the proof, not the hours.

The twelve months

Months	Focus	Hours per week	Milestone	Assessment
1–2	Foundation: Excel, SQL basics, basic statistics, business analytics basics	About 10	A cleaned dataset with a five-measure summary	A timed SQL test and an Excel cleaning exercise
3–4	Business intelligence: Power BI, star schema, DAX, dashboard design	About 10	A published dashboard with row-level security	Dashboard review against a rubric
5–6	Programming and data: Python, pandas, visualisation, Git and GitHub	About 12	A Python exploratory analysis of a public dataset in GitHub	Code review with tests
7–8	Machine learning and statistics: supervised learning, evaluation, a churn model	About 12	A churn model with a business recommendation	Model evaluation and explanation
9–10	Data engineering and cloud: warehouse concepts, ETL and ELT, dbt, Airflow, one cloud	About 12	A three-layer pipeline with tests	Pipeline review and a failure drill
11	GenAI, agents and MCP: LLM APIs, RAG, agents, MCP	About 12	A RAG assistant with citations and an evaluation	Evaluation report
12	Portfolio and interviews: two published projects, CV, LinkedIn, mock interviews, readiness check	About 15	Three mock interviews and two published projects	Interview rubric

The hours are a guide. A week with fewer hours is fine if the milestone is still reached; it is better to extend a stage than to skip its assessment.

Why this order

- **Excel and SQL first** because every later skill reads and writes tables. A learner who cannot join two tables cannot build a dashboard, a model or a pipeline.
- **Business intelligence before programming** because the first useful output, a dashboard, needs no code and proves the learner can answer business questions.

- **Python before machine learning** because models are built with code, and the data preparation takes most of the effort.
- **Statistics before machine learning** because the evaluation of a model is statistics; without it, a high accuracy number means little.
- **Data engineering after analytics** because pipelines exist to feed the analysis; the learner understands what the pipeline must deliver before building it.
- **GenAI and agents near the end** because they depend on the data, retrieval and evaluation skills earlier, and they are where mistakes are most expensive without those skills.

::::tip Protect the assessments An assessment is the proof that the stage is complete. A learner who skips it has consumed the content without the skill. Book the assessment in the calendar before starting each stage. ::::

What each milestone proves

Milestone	Skill it proves
Cleaned dataset with five measures	Cleaning, checking and summarising data
Published dashboard with RLS	Modelling, measures, sharing and access control
Python analysis in GitHub	Code, version control and reproducible analysis
Churn model with recommendation	Modelling, evaluation and communicating a decision
Three-layer pipeline with tests	Engineering reliability and data quality
RAG assistant with evaluation	Retrieval, grounding and measuring AI quality
Two published projects and mock interviews	Ability to show work and explain it under questioning

What employers generally expect

Employers look for working projects with code and data, clear documentation, measured results, the ability to explain trade-offs, and a steady record of learning. A roadmap cannot promise a job or a salary, and this one does not. It can make a candidate's evidence clear and specific.

WATCH OUT **Expect the first role to be a stepping stone**

Entry-level roles in analytics and data are competitive, and job titles vary between employers. A first role that uses part of the skill set is a good result, and the portfolio is what moves the next step.

Adjusting the plan

- If a stage's assessment is failed, repeat the stage's milestone before moving on.
- If your time drops, extend each stage rather than dropping the portfolio work.
- If you already know one stage, take its assessment early and move on.
- Review the roadmap at the end of each stage: what took longer, what helped, and what to change.

INTERVIEW CORNER **How would you explain your learning plan to an interviewer?**

Name the stages in order and the milestone each produced: a dashboard, a model, a pipeline, a RAG assistant. Explain why the order matters, show the two published projects, and say what you would learn next. Interviewers listen for a clear plan with evidence, not a long list of courses.

::::key What to remember - A realistic roadmap has weekly hours, a milestone and an assessment for each stage. - The order follows dependencies: data first, then analysis, programming, models, pipelines and AI. - Each milestone proves a specific skill; the portfolio makes the proof visible. - Employers look for evidence, clear documentation and explained trade-offs. - Extend a stage rather than skipping its assessment. ::::

Quick quiz

5 questions · tap an answer to check it

1. Why is business intelligence placed before programming in the roadmap?

A Power BI is easier than Python

B A dashboard proves business question skills without requiring code, and gives an early useful output

C Python is not needed

D It is the newest skill

Answer: B. The early milestone builds motivation and evidence.

2. Why should statistics come before machine learning?

A Statistics is required by law

B Evaluating a model is statistics, and a high accuracy number is meaningless without it

C Machine learning does not use data

D Statistics is shorter

Answer: B. Evaluation depends on statistical reasoning.

3. What should a learner do if an assessment is failed?

A Skip it and move on

B Repeat the stage's milestone before moving on

C Stop learning

D Change careers

Answer: B. The assessment proves the skill; repeat until it is shown.

4. Which milestone best proves interview readiness?

A A certificate

B Two published projects with mock interviews completed

C A list of videos watched

D A social media post

Answer: B. Published work and practised explanations are the evidence.

5. What does the roadmap promise?

A A guaranteed job

B Clear, specific evidence of skills, with no promised outcome

C A salary

D Certification

Answer: B. The roadmap cannot promise employment.

EXERCISE **Make your own year**

Adapt the roadmap to your situation: your weekly hours, your starting point, and the role you want. Write each stage with its months, hours, milestone and assessment. Then mark the two stages you would most likely skip under pressure and write how you will protect their assessments.

Show solution (worked answer)

Hints: start from the milestones, not the topics. If you have fewer hours, double the length of each stage rather than halving the content. Put the assessments in your calendar with dates.

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.